ADAN8888 — Week 9: Select the Winning Model

===== WEEK 9 START =====

Task: Compare all 9 previously trained configurations from Weeks 6–8 on the **same validation partition**. Lock the lowest-validation-MAE winner; build the bias–variance diagnostic; then evaluate **only that already-selected model** against the untouched final test period.

Crucial: Weeks 6–8 intentionally did not evaluate test data; Week 9 explicitly requires the final test evaluation.

1. Load the reusable Week 9 audit module

The notebook delegates to a single source implementation to prevent a separate, inconsistent modeling recipe.

In [3]:
```
from pathlib import Path
import importlib.util
import sys
import json
import numpy as np
from IPython.display import display, Image

cwd = Path.cwd().resolve()
ROOT = next((p for p in (cwd, *cwd.parents) if (p / 'Src' / 'week9_src_select_and_final_test.py').is_file()), None)
if ROOT is None:
    raise FileNotFoundError('Open the existing ADAN8888 repository in VS Code.')
source_file = ROOT / 'Src' / 'week9_src_select_and_final_test.py'
spec = importlib.util.spec_from_file_location('week9_model_audit', source_file)
week9 = importlib.util.module_from_spec(spec)
sys.modules[spec.name] = week9
spec.loader.exec_module(week9)
OUT = ROOT / 'Reports' / 'week9'
TABLES = OUT / 'week9_tables'
FIGURES = OUT / 'week9_figures'
print('Source module: Src/week9_src_select_and_final_test.py')
print('Historical validation sources: Reports/week6, Reports/week7, Reports/week8')
```

```
Source module: Src/week9_src_select_and_final_test.py
Historical validation sources: Reports/week6, Reports/week7, Reports/week8
```

2. Audit and rank all nine models using validation data only

Week 6: Ridge_A/Ridge_B/Ridge_C; Week 7: HGB_A/HGB_B/HGB_C; Week 8: MLP_A/MLP_B/MLP_C. The source prefers the original saved Week 6–8 metric CSVs; otherwise it uses the provided verified reference values (the Week 6 PDF was rounded to three decimal places).

Selection criterion: minimum validation MAE, followed by validation RMSE and then a declared heuristic complexity order for exact ties. No test labels are consulted in this step. The source writes `week9_selection_lock.json` before any test scoring.

In [4]:
```
selection = week9.select_and_lock(ROOT)
display(selection.comparison[['Model','Family','Configuration','Training_MAE','Validation_MAE','Training_RMSE','Validation_RMS
print('Validation ranking (best to worst):')
display(selection.ranking[['Model','Validation_MAE','Validation_RMSE']].round(3))
print('LOCKED winner:', selection.winner)
assert len(selection.comparison) == 9
assert len(selection.ranking) == 9
assert selection.winner == 'MLP_B'  # Expected from verified Week 6-8 results
assert selection.lock['TestDataViewedForSelection'] is False
assert selection.lock_path.is_file()
```

	Model	Family	Configuration	Training_MAE	Validation_MAE	Training_RMSE	Validation_RMSE	Training_R2	Validation_R2
0	Ridge_C	Ridge	alpha=1000000	3395.138	3418.394	5985.650	6163.435	0.405	0.422
1	Ridge_B	Ridge	alpha=10000	3332.425	3259.035	5789.185	5890.276	0.443	0.473
2	Ridge_A	Ridge	alpha=1	3331.959	3257.177	5787.133	5887.098	0.444	0.473
3	HGB_A	HGB	lr=0.05; iter=100; leaves=15	1180.219	1179.315	2263.184	2060.216	0.915	0.935
4	HGB_B	HGB	lr=0.05; iter=250; leaves=15	1164.083	1163.470	2254.863	2050.414	0.916	0.936
5	HGB_C	HGB	lr=0.05; iter=250; leaves=31	1159.017	1161.068	2247.904	2051.788	0.916	0.936
6	MLP_A	MLP	hidden=32-16; parameters=641	1171.375	1171.512	2236.708	2019.431	0.917	0.938
7	MLP_B	MLP	hidden=64-32; parameters=2305	1161.628	1143.195	2236.203	1988.778	0.917	0.940
8	MLP_C	MLP	hidden=64-32-16; parameters=2817	1159.014	1149.899	2230.234	1999.785	0.917	0.939

```
Validation ranking (best to worst):
```

	Model	Validation_MAE	Validation_RMSE
0	MLP_B	1143.195	1988.778
1	MLP_C	1149.899	1999.785
2	HGB_C	1161.068	2051.788
3	HGB_B	1163.470	2050.414
4	MLP_A	1171.512	2019.431
5	HGB_A	1179.315	2060.216
6	Ridge_A	3257.177	5887.098
7	Ridge_B	3259.035	5890.276
8	Ridge_C	3418.394	6163.435

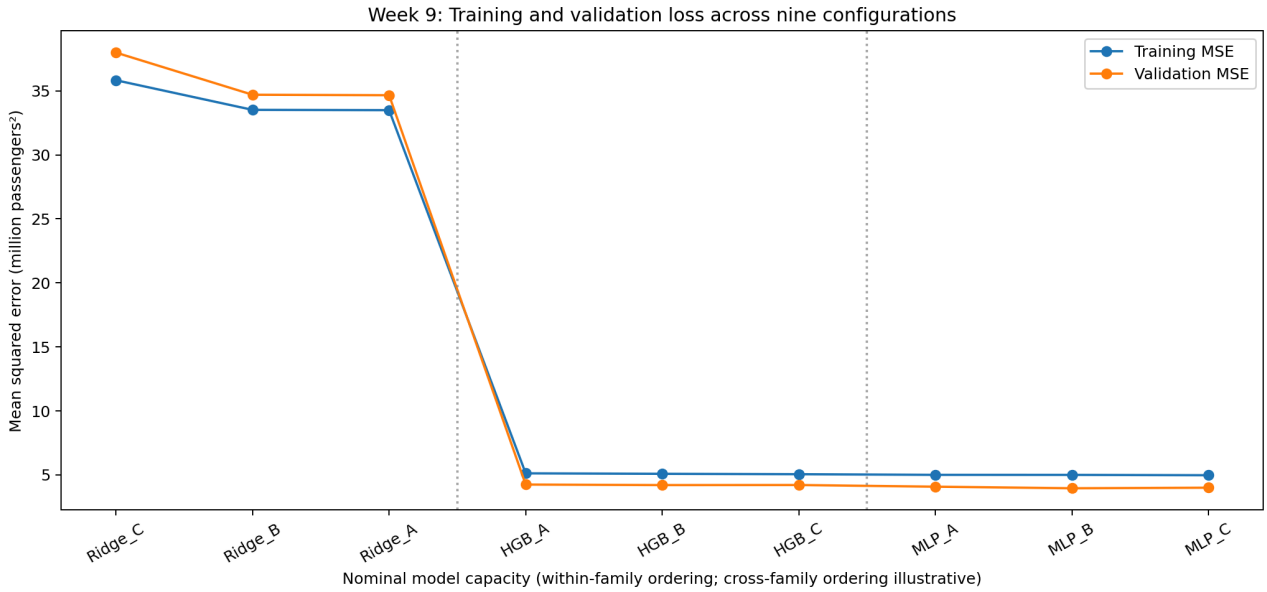
```
LOCKED winner: MLP_B
```

3. Bias–variance diagnostic across nine configurations

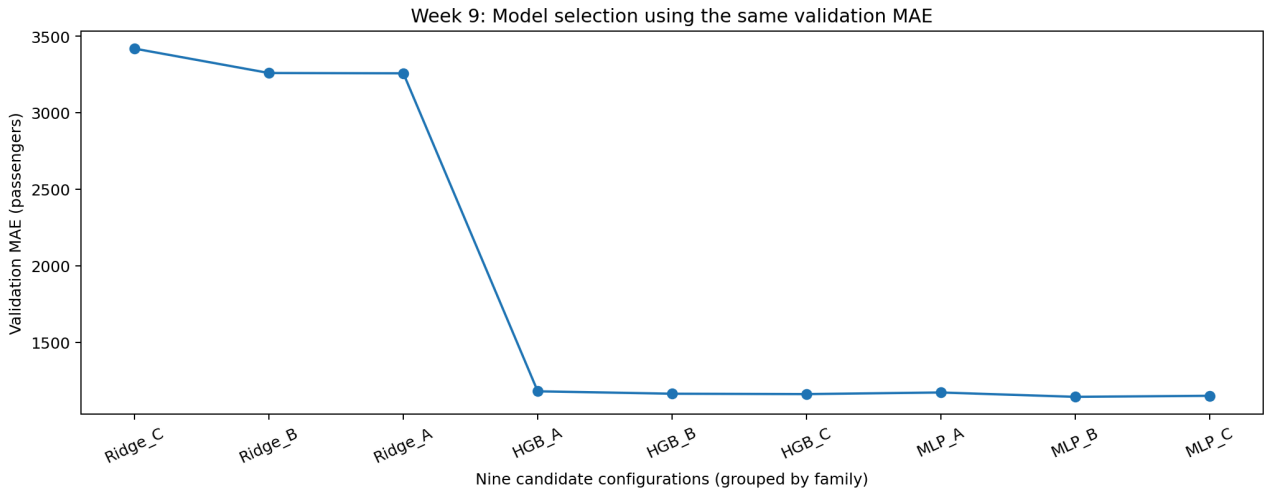
Interpretation caution: these are **observed training and validation mean squared errors**. Within Ridge, increasing L2 regularization reduces effective capacity; within HGB/MLP, more iterations, leaves, layers or neurons tend to increase capacity. **The relative complexity ranking across different model families cannot be measured on a single universal scale.** The plotted x-axis is a course-required, approximate/illustrative progression, not a mathematically proven ordering.

In [5]:
```
for filename in ['week9_figure_nine_model_bias_variance_diagnostic.png',
                 'week9_figure_nine_model_validation_mae.png']:
    assert (FIGURES / filename).exists()
    print(filename)
    display(Image(filename=str(FIGURES / filename)))
```

```
week9_figure_nine_model_bias_variance_diagnostic.png
```



week9_figure_nine_model_validation_mae.png



4. Final test: evaluate the locked, previously fitted MLP_B **once**

The saved model must exist at `Models/week8/week8_selected_mlp_neural_network.joblib`. The original Week 5 pipeline is recreated on the full official BTS data to obtain the same PC1/PC2 projection. The code **first reproduces MLP_B's historical validation MAE and RMSE within tight tolerance**. If either differs materially, the code refuses to evaluate the test set, preventing a mismatched model/preprocessing evaluation.

Only after this validation gate passes does the preselected model predict the final 264,291 test rows. The result and test plots are saved. **If a finalized test summary already exists, the notebook will reuse it instead of scoring test again.**

In [6]:
```
test_result = week9.final_test_once(ROOT, selection)
week9.write_report_values(ROOT, selection, test_result)
print('Selected model:', test_result['SelectedModel'])
print('Final test MAE (passengers):', f"{test_result['FinalTestMAE']:,.2f}")
print('Final test RMSE (passengers):', f"{test_result['FinalTestRMSE']:,.2f}")
print('Final test R²:', f"{test_result['FinalTestR2']:.4f}")
print('Validation-reproduction audit passed:', test_result['ValidationConsistencyCheckPassed'])
assert test_result['SelectedModel'] == selection.winner
assert test_result['TestLabelsUsedToSelectModel'] is False
assert test_result['RefitOnTestData'] is False
```

```
Selected model: MLP_B
Final test MAE (passengers): 1,122.50
Final test RMSE (passengers): 1,935.24
Final test R²: 0.9390
Validation-reproduction audit passed: True
```

5. Final winning model: training, validation and held-out test metrics

The training and validation metrics are imported unchanged from Week 8's actual model comparison. The test metrics are computed using the frozen, prior-training-fitted MLP_B model. The test period is not used to make new model/parameter decisions.

In [7]:
```
from pandas import read_csv
winner_table = read_csv(TABLES / 'week9_winner_train_validation_test_metrics.csv')
display(winner_table.round(4))
assert list(winner_table.Partition) == ['Training','Validation','Test']
assert len(winner_table) == 3
assert winner_table.loc[winner_table.Partition.eq('Test'),'Rows'].iloc[0] == 264291
```

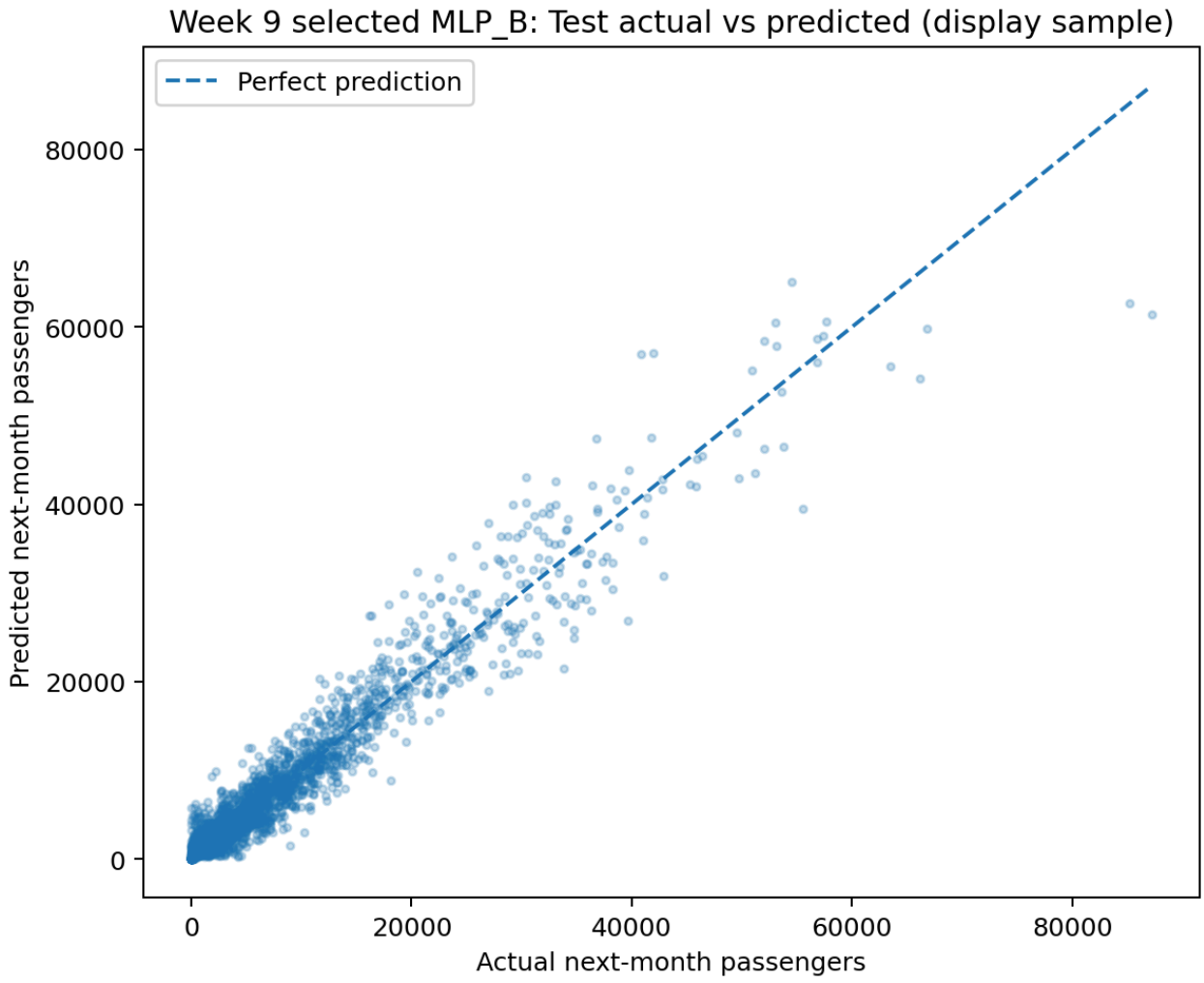
	Partition	Rows	MAE	RMSE	R2
0	Training	1062603	1161.6278	2236.2033	0.9169
1	Validation	239395	1143.1952	1988.7780	0.9399
2	Test	264291	1122.5049	1935.2387	0.9390

6. Inspect final test diagnostic figures

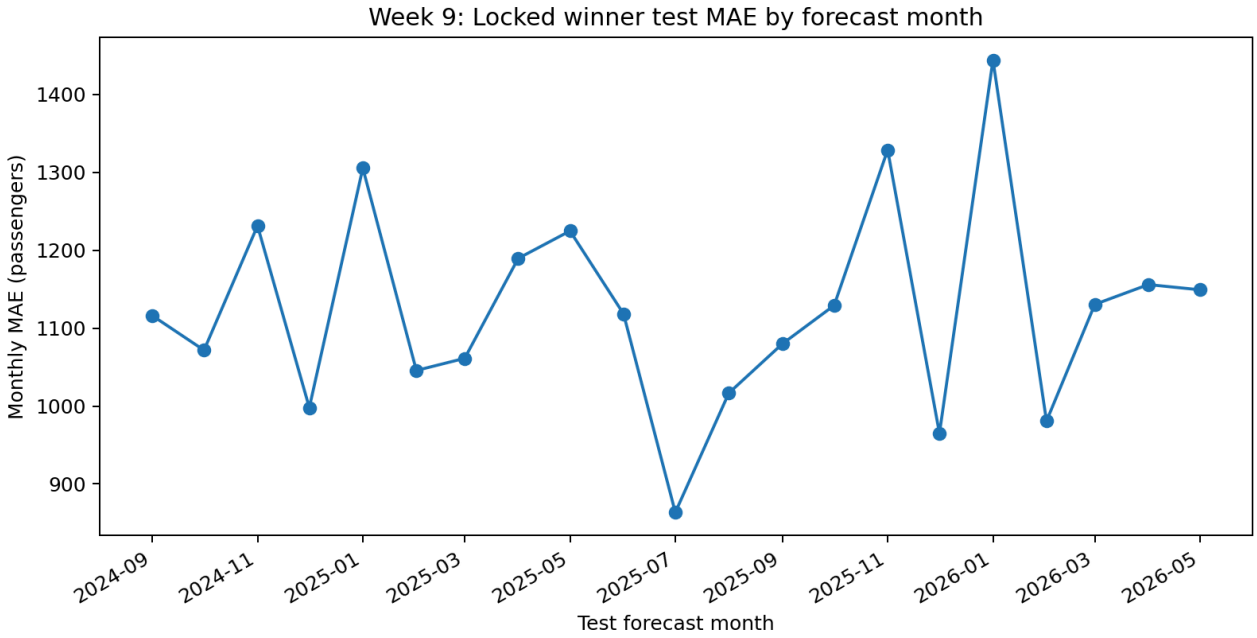
The actual-versus-predicted scatter uses at most 4,000 points *for visualization only*. All reported metrics and monthly errors use the **full test partition**.

In [8]:
```
for filename in ['week9_figure_test_actual_vs_predicted.png','week9_figure_test_monthly_mae.png']:
    assert (FIGURES / filename).is_file(), filename
    print(filename)
    display(Image(filename=str(FIGURES / filename)))
```

week9_figure_test_actual_vs_predicted.png



week9_figure_test_monthly_mae.png



7. Final checks, reproducibility and report input

The integrity checks below verify that the nine historical scores, the fixed selection decision, and the single test-result record are available. The final written report should compare all nine validation results; discuss the training–validation complexity chart; and answer whether final **test** MAE/RMSE/R^2 are adequate for deployment. That last judgment requires an explicit airline planning tolerance or comparison to a business baseline; a good validation score alone does not establish readiness.

Note: This is a retrospective held-out test of an already-fitted model. It is not a deployment benchmark for forecasts made before BTS data are published.

In [9]:
```
assert (TABLES / 'week9_nine_model_comparison.csv').is_file()
assert (TABLES / 'week9_validation_ranking.csv').is_file()
assert (TABLES / 'week9_selection_lock.json').is_file()
assert (TABLES / 'week9_test_winner_summary.json').is_file()
assert (TABLES / 'week9_winner_train_validation_test_metrics.csv').is_file()
assert (OUT / 'week9_verified_results_for_report.md').is_file()
assert len(selection.comparison) == 9
assert test_result['TestEvaluatedAfterSelectionLock'] is True
assert np.isfinite([test_result['FinalTestMAE'],test_result['FinalTestRMSE'],test_result['FinalTestR2']]).all()
print('All Week 9 checks passed.')
print('===== WEEK 9 END =====')
```

```
All Week 9 checks passed.
===== WEEK 9 END =====
```